# Computer Forensics and cybercrime analysis

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*(many Images created with flux-1 text to image)*

# the topic

- **forensic analysis (forensic science)**
  - the application of scientific knowledge and methodology to legal problems and criminal investigations
- **computer forensics**
  - *systematic collection, preservation, analysis, and presentation of digital evidence to support investigations and legal proceedings*

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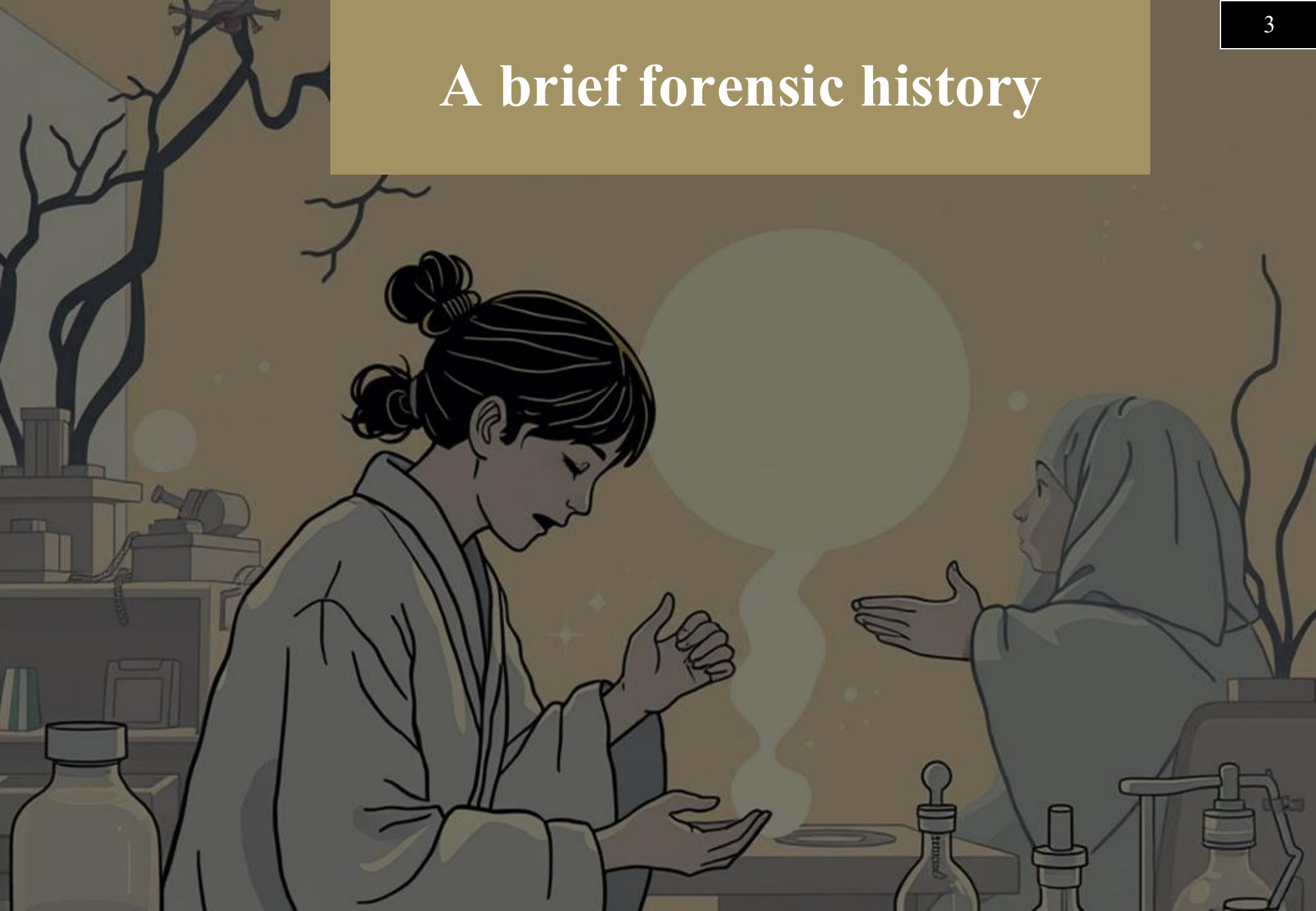
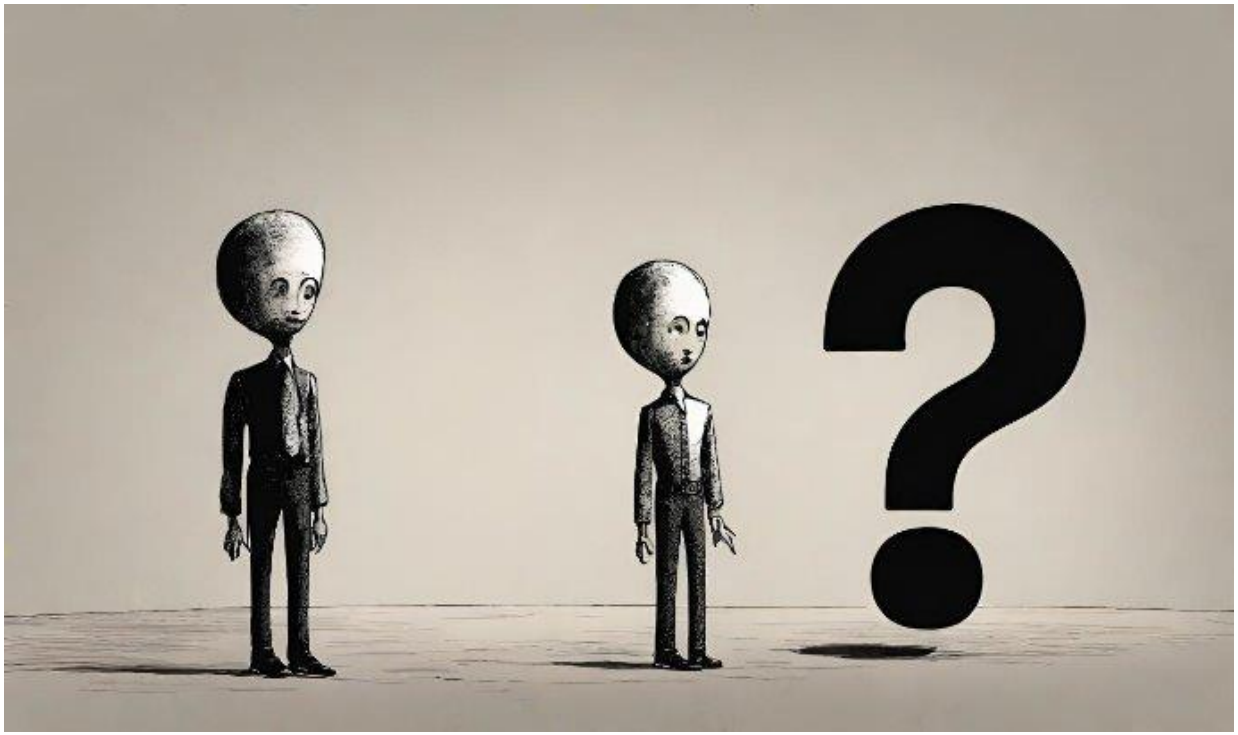
# A brief forensic history



- ***How old is forensics science (loosely speaking)?***



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# Forensic science history - fast forward (I)

- **Forensics science – ancient times**
  - fingerprints for identification (Babylon, ~1900s BC, business)
  - *Hsi Duan Yu* ("The washing away of wrongs", 1248 AC, China) introduces forensic pathology (*drowning* vs *strangulation*)
  - in 1835 first Bullet comparison used in a murder investigation (UK, Henry Goddard, member of British Police Force)
  - 1880s
    - Henry Faulds proposed ink and fingerprint for people identification
    - Sir Francis Galton collected thousands fingerprints and developed a classification based on features like spirals, loops, arches
  - first use of fingerprints ("*comparative dactyloscopy*") to solve a murder in 1892 (Juan Vucetich –Argentina)

# Forensic science history - fast forward (II)

- **Forensics science – "modern" times**
  - August Vollmer (Los Angeles Police Chief) established the first crime lab (followed three years later by the forerunner of FBI) in 1923
  - development of DNA fingerprinting (officially dated in 1950s)
    - DNA profiling developed by Sir Alec Jeffreys - (from) 1985
  - late 80s – early 90s: Automated fingerprint identification systems (AFIS) development and adoption
    - "*The system, operated by the State Police, is designed to improve the efficiency and effectiveness of criminal investigations by classifying, matching, storing, and retrieving fingerprints*."
  - present: integration of AI, forensic toxicology, and digital forensics into modern crime-solving

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# Digital Forensic science history (I)

- **Forensics science – digital field – early times**
  - United States v. Robert Tappan Morris (1989)
    - first person convicted under the Computer Fraud and Abuse Act after releasing the Morris Worm
    - digital forensics adopted to analyze computer logs and network activity, for the first time to such extent
  - founding of the International Association of Computer Investigative Specialists (IACIS - 1989)
    - "*developing guidelines for the seizure and processing of computer evidence" (*and business cards, logos, etc*)*
  - Int. Organization on Computer Evidence (IOCE) - Founded in 1995
    - forum for law enforcement agencies to exchange information computer investigation and computer forensic issue

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# Digital Forensic science history (II)

- **Forensics science – digital field – recent times**
  - 1990s: Emergence of basic forensic tools like EnCase.
  - 2000s: Widespread adoption of digital forensics in law enforcement
  - 2010s: Growth of cloud and mobile forensics automation and machine learning to support forensic investigations
  - 2020s: Advanced encryption and blockchain forensics, adoption of LLMs to improve digital forensics effectiveness, evolution of cybercrime frameworks and international collaborations

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# Definitions

# Computer Forensics definitions

More on Prof. Vaciago' lessons

The discipline that combines elements of law and computer science to collect and analyze data from computer systems, networks, wireless communications, and storage devices in a way that is admissible as evidence in a court of law.

*US_CERT*

The discipline whose goal is preservation, identification, analysis of information system to the aim of identification of evidences during investigation activities

*A. Ghirardini -Computer Forensics*

The application of computer science and investigative procedures involving the examination of digital evidence - following proper search authority, chain of custody, validation with mathematics, use of validated tools, repeatability, reporting, and possibly expert testimony

*NIST glossary*

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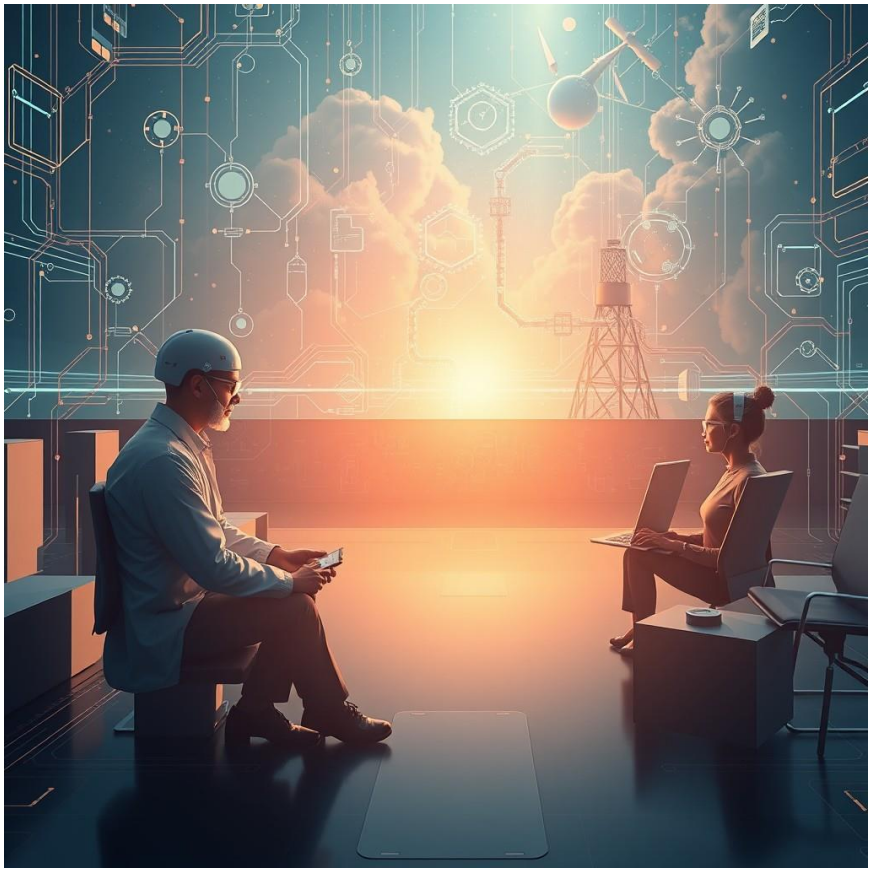
# CF purpose(s)



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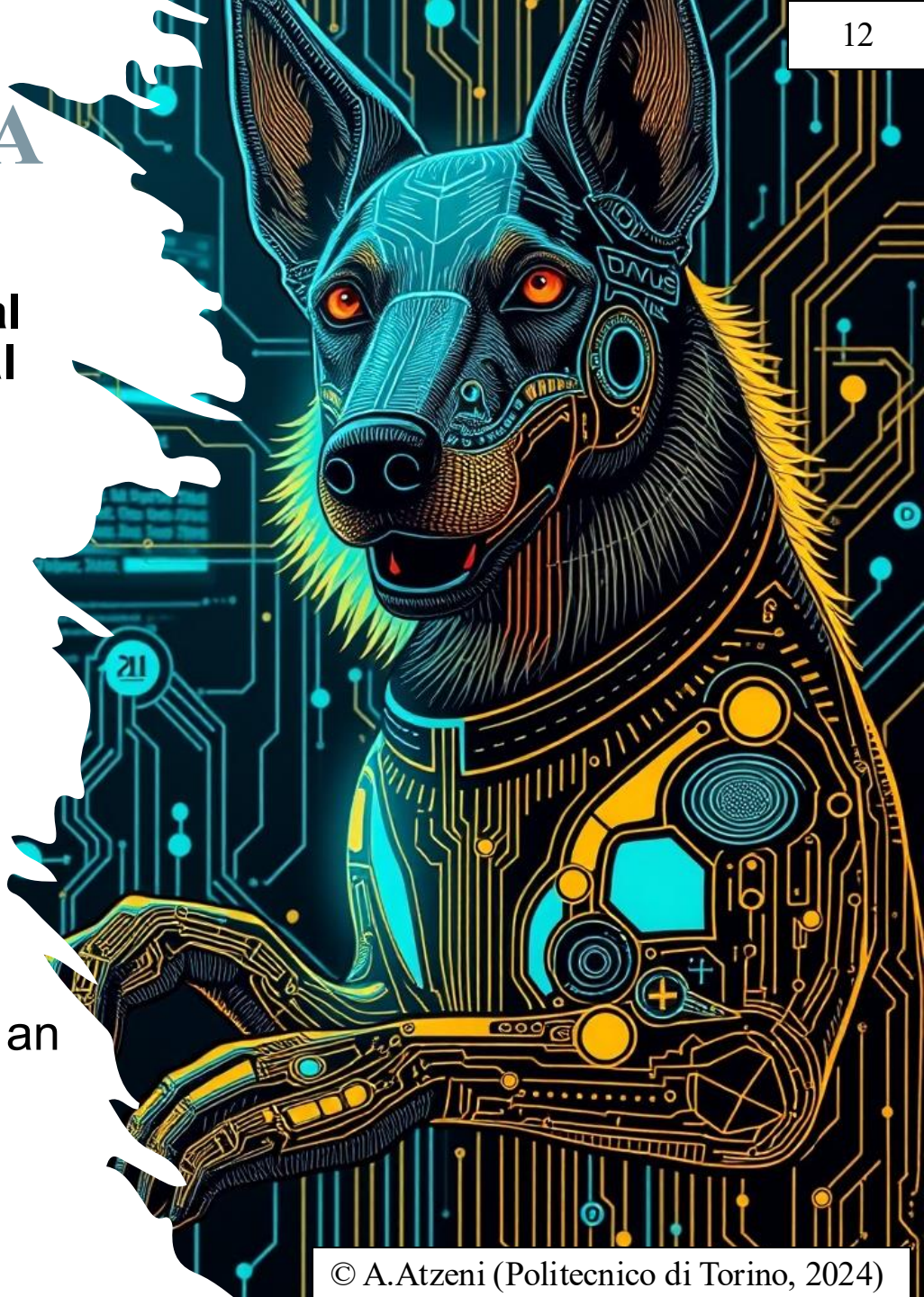
# Computer Forensics Q&A

- **digital forensics seeks to uncover, analyze, and interpret data from digital devices to answer a few foundamental key questions**
  - what happened?
  - who was involved?
  - when did it take place?
  - where did it take place?
  - why did it take place?
  - how did an incident occur?
- **whose answers allow to**
  - support legal proceedings
    - i.e. confirm or refute allegations of an incident
  - support the mitigation of damages
  - mature future prevention approaches



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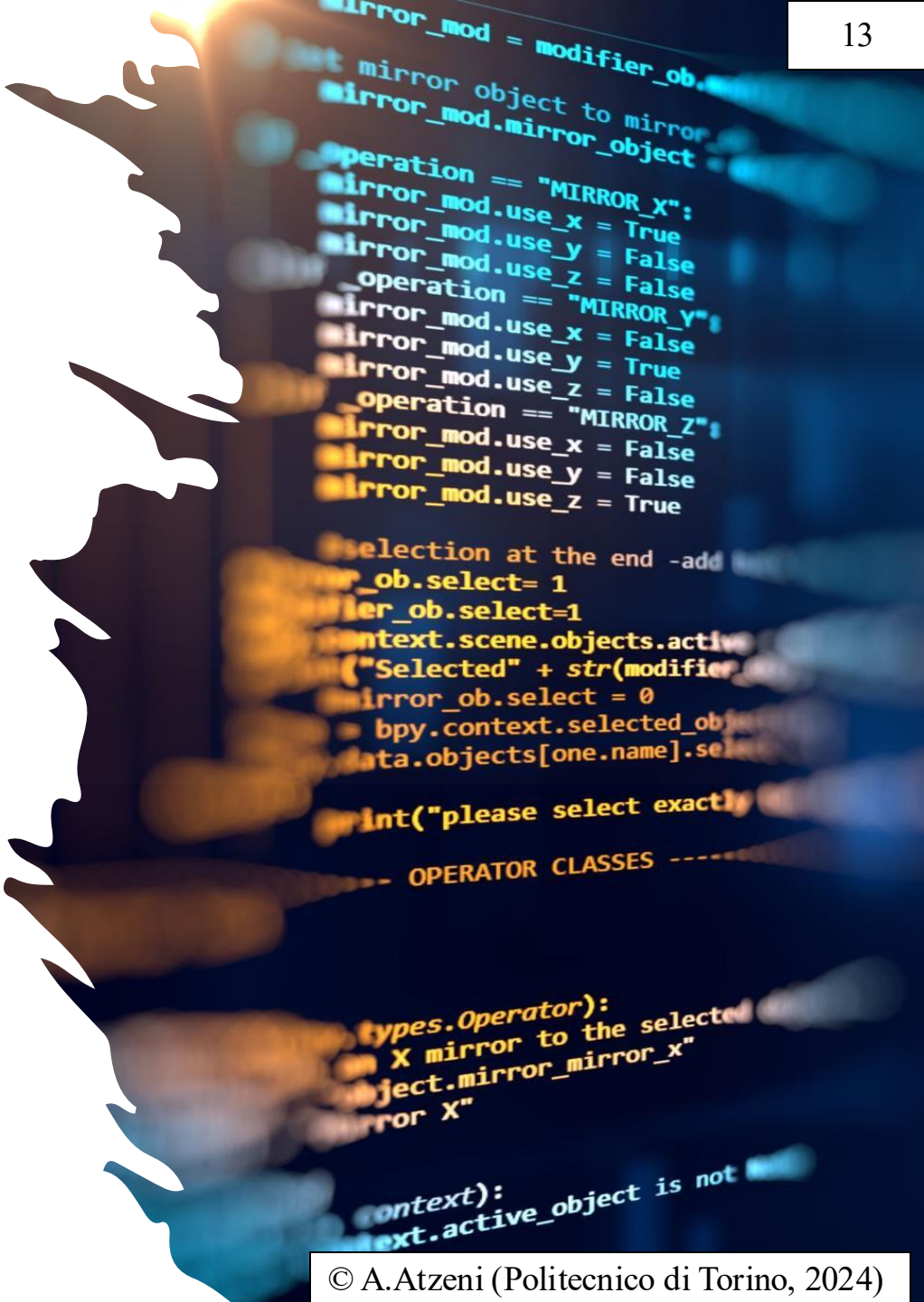
# Computer Forensics technical goals

- **retrieve what has been the input**
  - e.g. what has been typed
- **determine actions performed**
  - e.g. what program have been run, what peripherals have been connected
- **analyse used files**
  - e.g. what modifications have been done, when modifications occurred
- **identify damage done**
  - e.g. what data have been erased
- **In essence**, ***gain comprehension of what happened (technically speaking)***

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